

Systems Biology: Mathematics for Biologists



Levien van Zon, Theoretical Biology, UU

Introductie: Wiskunde

CIJFERLIJST

EINDEXAMEN VOORBEREIDEND WETENSCHAPPELIJK ONDERWIJS

Afgelegd aan: **FSG Krommenie - Bertrand Russell**
Krommenie

vak/vakken waarin examen is afgelegd ¹⁾	cijfers toegekend voor het:		EINDCIJFERS	
	school- onderzoek	centraal examen	in cijfers	in letters
1. Nederlandse taal en letterkunde	8.4	7.4	8	acht
2. Engelse taal en letterkunde	8.5	9.3	9	negen
3. geschiedenis en staatsinrichting	8.3	7.0	8	acht
4. wiskunde B	5.1	4.0	5	vijf
5. natuurkunde	7.3	8.0	8	acht
6. scheikunde	7.2	8.9	8	acht
7. biologie	8.4	7.0	8	acht
8. xxx				
9. xxx				
10. xxx				

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10. xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx				

When Math Hurts: Math Anxiety Predicts Pain Network Activation in Anticipation of Doing Math

Ian M. Lyons^{1,2*}, Sian L. Beilock^{1*}

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Abstract

Math can be difficult, and for those with high levels of mathematics-anxiety (HMAs), math is associated with tension, apprehension, and fear. But what underlies the feelings of dread effected by math anxiety? Are HMAs' feelings about math merely psychological epiphenomena, or is their anxiety grounded in simulation of a concrete, visceral sensation – such as pain – about which they have every right to feel anxious? We show that, when anticipating an upcoming math-task, the higher one's math anxiety, the more one increases activity in regions associated with visceral threat detection, and often the experience of pain itself (bilateral dorso-posterior insula). Interestingly, this relation was not seen during math performance, suggesting that it is not that math itself hurts; rather, the anticipation of math is painful. Our data suggest that pain network activation underlies the intuition that simply anticipating a dreaded event can feel painful. These results may also provide a potential neural mechanism to explain why HMAs tend to avoid math and math-related situations, which in turn can bias HMAs away from taking math classes or even entire math-related career paths.

Citation: Lyons IM, Beilock SL (2012) When Math Hurts: Math Anxiety Predicts Pain Network Activation in Anticipation of Doing Math. PLoS ONE 7(10): e48076. doi:10.1371/journal.pone.0048076

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Introductie: Wiskunde

Vandaag: **Lineaire algebra** en **complexe getallen**

Nodig voor onderdelen Wiskunde en Theoretische Biologie
(1 maart t/m 8 april, tentamen 12 april).

- Thuis: Reader lezen (vóór college), eventueel opgaven maken.
 - Werkcollege: Opgaven maken (oefenen!) en vooral ook vragen stellen.
 - Gezamenlijke bonuspunt Wiskunde en Theoretische Biologie.
 - Alle antwoorden staan achterin het dictaat.
 - Extra materiaal: <http://www-binf.bio.uu.nl/sb/wiskunde.html>
-
- Er bestaan geen domme vragen, we willen je graag helpen!
 - Doel is niet om jullie zo moeilijk mogelijke wiskunde te leren.
 - Doel is begrip, en laten zien dat wiskunde nodig is voor biologische inzichten.
 - Onderwerpen vaak abstract, maar nodig om later in de cursus dingen te begrijpen.

Vectors, Matrices and Complex numbers

Contents

- Scalars, vectors, matrices.
 - Adding and multiplying vectors.
 - Adding and multiplying matrices.
 - Matrix transformations (rotation, scaling, shearing).
 - Matrix determinant.
- Linear systems.
- Eigenvectors, eigenvalues.
 - What are they?
 - Determining eigenvalues.
 - Determining eigenvectors.
- Complex numbers
 - What are they?
 - How to use them.
 - Adding, multiplying, dividing.

Scalars, vectors and matrices

Scalar

single number: $\lambda = 5$

e.g. expression level of a gene

Vector

column or row of n numbers: $\vec{V} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ or $\vec{V} = (2 \ 1)$

e.g. expression levels of two genes

Matrix

block of n rows and m columns of numbers: $A = \begin{pmatrix} 2 & 4 & 3 \\ 1 & 7 & 4 \end{pmatrix}$

e.g. expression levels of two genes in 3 experiments

Scalar arithmetic

We'll assume you are familiar with this:

Addition

$a + b = b + a = c$, e.g. $5 + 4 = 9$ and $4 + 5 = 9$

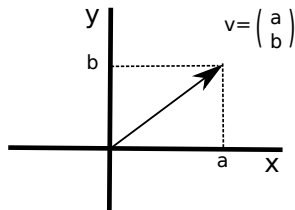
Multiplication

$a \times b = b \times a = d$, e.g. $5 \times 4 = 20$ and $4 \times 5 = 20$

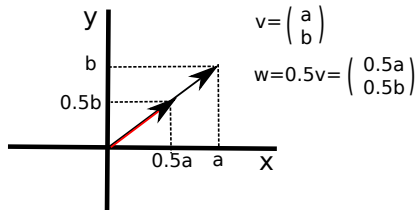
This is also true for vectors, but different for matrices!

Vector arithmetic

A vector in 2 dimensions



Scaling of vector



Length

$$|\vec{v}| = \sqrt{a^2 + b^2 + \dots n^2}$$

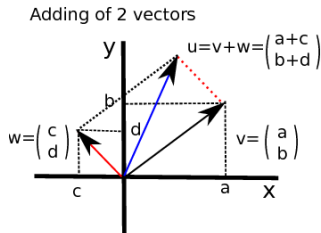
e.g. $\vec{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ so $|\vec{v}| = \sqrt{2^2 + 1^2} = \sqrt{4 + 1} = \sqrt{5}$

Scaling

$$a \times \vec{v} = \vec{u}, \text{ with } u_i = a \times v_i$$

e.g. $3 \times \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 3 \end{pmatrix}$ or $0.5 \times \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0.5 \end{pmatrix}$

Vector arithmetic



Addition

$$\vec{v} + \vec{w} = \vec{w} + \vec{v} = \vec{u}, \text{ with } u_i = v_i + w_i$$

$$\text{e.g. } \begin{pmatrix} -1 \\ 2 \end{pmatrix} + \begin{pmatrix} 4 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix} \text{ and } \begin{pmatrix} 4 \\ 3 \end{pmatrix} + \begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$

Vector arithmetic

Multiplication (dot product)

$$\vec{v} \times \vec{w} = \vec{w} \times \vec{v} = a, \text{ with } a = \sum v_i \times w_i$$

$$\text{e.g. } \begin{pmatrix} -1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 3 \end{pmatrix} = -1 \times 4 + 2 \times 3 = 2$$

$$\begin{pmatrix} -1 & 2 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 3 \end{pmatrix} = 2$$

$$\begin{pmatrix} 4 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 2 \end{pmatrix} = 4 \times -1 + 3 \times 2 = 2$$

Matrix arithmetic

Scaling

$$c \times A = B, \text{ with } B_{i,j} = c \times A_{i,j}$$

$$\text{e.g. } 4 \times \begin{pmatrix} 1 & 4 & 5 \\ 2 & 6 & 10 \end{pmatrix} = \begin{pmatrix} 4 & 16 & 20 \\ 8 & 24 & 40 \end{pmatrix}$$

Addition

$$A + B = B + A = C, \text{ with } c_{i,j} = a_{i,j} + b_{i,j}, \text{ only if A en B same sizes}$$

$$\text{e.g. } \begin{pmatrix} 1 & 4 & 5 \\ 2 & 6 & 10 \end{pmatrix} + \begin{pmatrix} 2 & 1 & 4 \\ 1 & 3 & 5 \end{pmatrix} = \begin{pmatrix} 3 & 5 & 9 \\ 3 & 9 & 15 \end{pmatrix}$$

Matrix arithmetic

Multiplication

$$A \times B = C, \text{ with } c_{ij} = \sum_{k=1}^p a_{ik} b_{kj}, \text{ if no. columns } A = \text{no. rows } B = p$$

$$\begin{pmatrix} -1 & 2 \\ 4 & 1 \end{pmatrix} \times \begin{pmatrix} 4 & 1 \\ 3 & 5 \end{pmatrix} = \begin{pmatrix} -1 \cdot 4 + 2 \cdot 3 & -1 \cdot 1 + 2 \cdot 5 \\ 4 \cdot 4 + 1 \cdot 3 & 4 \cdot 1 + 1 \cdot 5 \end{pmatrix} = \begin{pmatrix} 2 & 9 \\ 19 & 9 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 2 \\ 4 & 1 \end{pmatrix} \times \begin{pmatrix} 4 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 19 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 2 \end{pmatrix} \times \begin{pmatrix} 4 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \end{pmatrix}$$

$$B \times A = D \neq C, \text{ with } d_{ij} = \sum_{k=1}^p b_{ik} a_{kj}, \text{ if no. columns } B = \text{no. rows } A = p$$

$$\begin{pmatrix} 4 & 1 \\ 3 & 5 \end{pmatrix} \times \begin{pmatrix} -1 & 2 \\ 4 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 9 \\ 17 & 11 \end{pmatrix}$$

Matrices: Vector transformations

Transformation matrix

Multiplying a vector with a matrix transforms the vector:

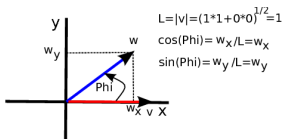
$$A \times \vec{v} = \vec{w}$$

Usually the new vector has a new direction or length, or both.

Rotation matrix

$$A\vec{v} = \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} v_x \cos \phi - v_y \sin \phi \\ v_x \sin \phi + v_y \cos \phi \end{pmatrix}$$

The new vector has the same length but a different direction.

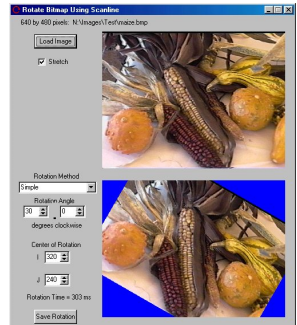
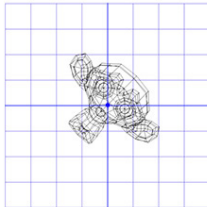
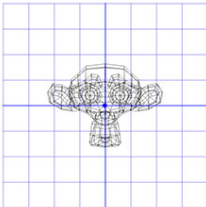


Matrices: Vector rotation

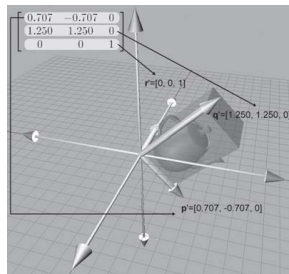
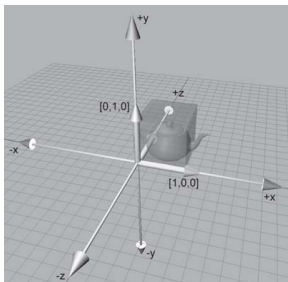
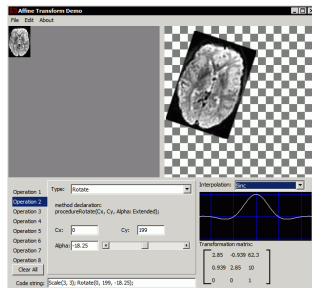
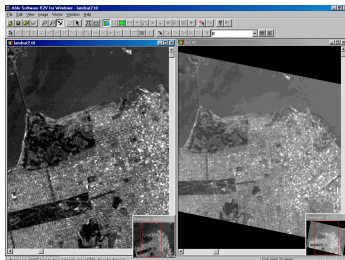


MATRIX TRANSFORM

$$\begin{bmatrix} \cos 90^\circ & \sin 90^\circ \\ -\sin 90^\circ & \cos 90^\circ \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$



Matrices: Vector rotation

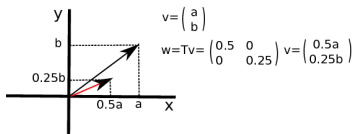


Matrices: More vector transformations

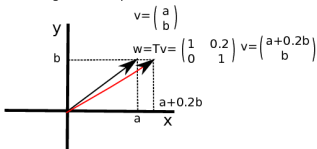
Scaling and shearing matrices

Other matrix transformations of vectors:

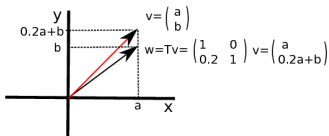
Complex scaling of vector



Shearing of vector parallel to x-axis



Shearing of vector parallel to y-axis



New vectors have different lengths and directions.

Matrices: Shearing transformations

Original image



Image sheared in x direction



Shear Matrix = $\begin{bmatrix} 1 & 0.5 & 0 \\ 0 & 1 & 0 \end{bmatrix}$



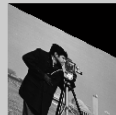
Shear Matrix = $\begin{bmatrix} 1 & 0 & 0.5 \\ 0 & 1 & 0 \end{bmatrix}$



Shear Matrix = $\begin{bmatrix} 1 & -0.5 & 0 \\ 0 & 1 & 0 \end{bmatrix}$



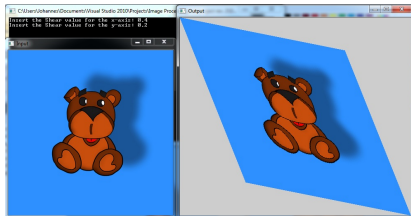
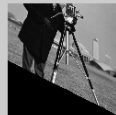
Shear Matrix = $\begin{bmatrix} 1 & 0 & -0.5 \\ 0 & 1 & 0 \end{bmatrix}$



FLIPUD and
Shear Matrix = $\begin{bmatrix} 1 & 0.5 & 0 \\ 0 & 1 & 0 \end{bmatrix}$



FLUPLR and
Shear Matrix = $\begin{bmatrix} 1 & 0 & 0.5 \\ 0 & 1 & 0 \end{bmatrix}$



Matrices: Fish transformations



Papegaaivis (*Scarus*)



Keizersvis (*Pomacanthus*)

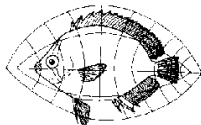
Matrices: Fish transformations

D'Arcy Thompson (1917)

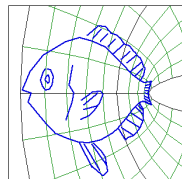
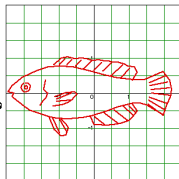
Modeling growth, development, evolution:



Scarus sp.



Pomacanthus.



Apply series of scaling, shearing and rotation transforms.

Note that $A \times B \neq B \times A$, so order of transforms matters.

Matrix determinant

Determinant

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\det(A) = \det \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - cb$$

$$\text{e.g. } \det \begin{vmatrix} -1 & 2 \\ 4 & 1 \end{vmatrix} = -1 \times 1 - 4 \times 2 = -9$$

(you only need to know $\det$ for 2×2 matrices in this course)

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 - Adding, multiplying, dividing.

Matrices, vectors and systems of linear equations

$$A \cdot \vec{v} = \vec{w}$$

Consider this:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} p \\ q \end{pmatrix}$$

$$\begin{cases} ax + by = p \\ cx + dy = q \end{cases}$$

$$\begin{pmatrix} -1 & 2 \\ 4 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} -x + 2y = 0 \\ 4x + y = 0 \end{cases}$$

Matrices, vectors and systems of linear equations

The system:

$$\begin{cases} ax + by = p \\ cx + dy = q \end{cases}$$

can be written as:

$$A\vec{X} = \vec{V}$$

with

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}; \vec{X} = \begin{pmatrix} x \\ y \end{pmatrix}; \vec{V} = \begin{pmatrix} p \\ q \end{pmatrix}$$

Eigenvalues and Eigenvectors

Usually

$$A\vec{v} = \vec{w}$$

Matrix A transforms vector $\vec{v}$ into different vector $\vec{w}$, with a different direction.

For special $\vec{v}$

$$A\mathbf{v} = \lambda\mathbf{v}$$

Matrix A behaves as a scalar.

Eigenvalues and Eigenvectors

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1+2 \\ 2+1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$A \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 3 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$2 \cdot \left(A \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right) = 2 \cdot \left(3 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right)$$

$$A \cdot \begin{pmatrix} 2 \\ 2 \end{pmatrix} = 3 \cdot \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

Eigenvalues and Eigenvectors

Usually

$$A\vec{v} = \vec{w}$$

Matrix A transforms vector $\vec{v}$ into different vector $\vec{w}$

For special $\vec{v}$ (not zero)

$$A\mathbf{v} = \lambda\mathbf{v}$$

Matrix A only rescales $\vec{v}$ into $\lambda\vec{v}$

We call $\mathbf{v}$ an **eigenvector**, and the factor λ by which it is rescaled its corresponding **eigenvalue**.

Eigenvalues and Eigenvectors

Non-uniqueness of eigenvectors

Eigenvectors can be scaled by an arbitrary constant k :

$$kA\mathbf{v} = k\lambda\mathbf{v} \quad \text{or} \quad A(k\mathbf{v}) = \lambda(k\mathbf{v})$$

$k\mathbf{v}$ is also an eigenvector corresponding to eigenvalue λ

Number of eigenvalues

An $n \times n$ square matrix has a maximum of n eigenvalues.

Eigenvalues and Eigenvectors

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 3 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1-2 \\ 2-1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} = -1 \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

The eigenvalues of matrix $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ are $\lambda_1 = 3$ and $\lambda_2 = -1$.

The eigenvectors are $\mathbf{v}_1 = k \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{v}_2 = k \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

Finding eigenvalues

$$A \cdot \mathbf{v} = \lambda \cdot \mathbf{v}$$

Can be written as:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \lambda \begin{pmatrix} v_x \\ v_y \end{pmatrix} \Rightarrow \begin{pmatrix} a - \lambda & b \\ c & d - \lambda \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Or:

$$\begin{cases} av_x + bv_y = \lambda v_x \\ cv_x + dv_y = \lambda v_y \end{cases} \Rightarrow \begin{cases} (a - \lambda)v_x + bv_y = 0 \\ cv_x + (d - \lambda)v_y = 0 \end{cases}$$

Gives the so-called **characteristic equation**:

$$\lambda^2 - (a + d)\lambda + (ad - cb) = 0$$

Which can be written as:

$$\det \begin{vmatrix} a - \lambda & b \\ c & d - \lambda \end{vmatrix} = (a - \lambda)(d - \lambda) - bc = 0$$

Finding eigenvalues

Example:

Find eigenvalues of:

$$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

$$\det \begin{vmatrix} 1 - \lambda & 2 \\ 2 & 1 - \lambda \end{vmatrix} = 0$$

$$(1 - \lambda)(1 - \lambda) - 2 \times 2 = 0$$

$$\lambda^2 - 2\lambda - 3 = 0$$

$$\lambda = 3 \vee \lambda = -1$$

Finding eigenvectors

Substitute eigenvalues ($\lambda_1 = 3$, $\lambda_2 = -1$) in:

$$\begin{pmatrix} a - \lambda & b \\ c & d - \lambda \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} (a - \lambda)v_x + bv_y = 0 \\ cv_x + (d - \lambda)v_y = 0 \end{cases}$$

For $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ and $\lambda_1 = 3$:

$$\begin{cases} (1 - 3)v_x + 2v_y = 0 \\ 2v_x + (1 - 3)v_y = 0 \end{cases} \Rightarrow \begin{cases} -2v_x + 2v_y = 0 \\ 2v_x - 2v_y = 0 \end{cases} \Rightarrow \begin{cases} -2v_x = -2v_y \\ 2v_x = 2v_y \end{cases}$$

so $v_x = v_y$, so corresponding eigenvector is $\mathbf{v}_1 = k \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Note that the first and second equation give the same info.

Express method for finding eigenvectors

$$\begin{pmatrix} a - \lambda & b \\ c & d - \lambda \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} (a - \lambda)v_x + bv_y = 0 \\ cv_x + (d - \lambda)v_y = 0 \end{cases}$$

First and second equation give the same info.

Using only the first equation we would get:

$$(a - \lambda)v_x + bv_y = 0$$

$$(a - \lambda)v_x = -bv_y$$

$$\begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} -b \\ a - \lambda \end{pmatrix}$$

Express method for finding eigenvectors

$$\begin{pmatrix} a - \lambda & b \\ c & d - \lambda \end{pmatrix} \begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} (a - \lambda)v_x + bv_y = 0 \\ cv_x + (d - \lambda)v_y = 0 \end{cases}$$

First and second equation give the same info.

Using only the second equation we would get:

$$cv_x + (d - \lambda)v_y = 0$$

$$cv_x = -(d - \lambda)v_y$$

$$cv_x = (-d + \lambda)v_y$$

$$\begin{pmatrix} v_x \\ v_y \end{pmatrix} = \begin{pmatrix} -d + \lambda \\ c \end{pmatrix}$$

Express method for finding eigenvectors

Finding the eigenvectors of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with eigenvalues λ_1, λ_2 .

Using the first equation:

$$\mathbf{v}_1 = \begin{pmatrix} v_{1x} \\ v_{1y} \end{pmatrix} = \begin{pmatrix} -b \\ a - \lambda_1 \end{pmatrix} \quad \mathbf{v}_2 = \begin{pmatrix} v_{2x} \\ v_{2y} \end{pmatrix} = \begin{pmatrix} -b \\ a - \lambda_2 \end{pmatrix}$$

Using the second equation:

$$\mathbf{v}_1 = \begin{pmatrix} v_{1x} \\ v_{1y} \end{pmatrix} = \begin{pmatrix} d - \lambda_1 \\ -c \end{pmatrix} \quad \mathbf{v}_2 = \begin{pmatrix} v_{2x} \\ v_{2y} \end{pmatrix} = \begin{pmatrix} d - \lambda_2 \\ -c \end{pmatrix}$$

If one of two methods gives a zero vector, use the other one!

Summary: Eigenvalues and Eigenvectors

Matrix transformation: $A\vec{v} = \vec{w}$

For special eigenvectors $\vec{v}$ (not zero): $A\mathbf{v} = \lambda\mathbf{v}$

The matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ has a maximum of 2 eigenvalues $\lambda_{1,2}$.

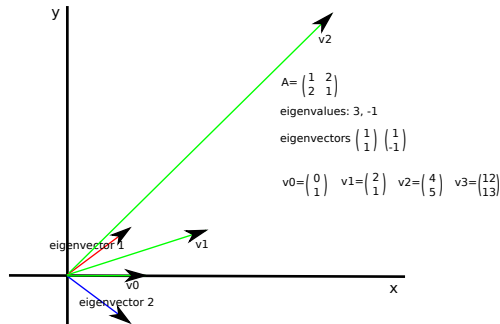
Finding the eigenvalues for a 2×2 matrix:

$$\det \begin{vmatrix} a - \lambda & b \\ c & d - \lambda \end{vmatrix} = (a - \lambda)(d - \lambda) - bc = 0$$

Finding the corresponding eigenvectors:

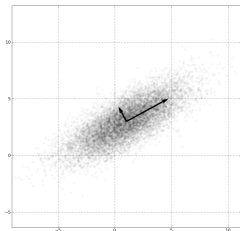
$$\mathbf{v}_{1,2} = \begin{pmatrix} -b \\ a - \lambda_{1,2} \end{pmatrix} \quad \text{or} \quad \mathbf{v}_{1,2} = \begin{pmatrix} -d + \lambda_{1,2} \\ c \end{pmatrix}$$

Eigenvectors as dominant directions

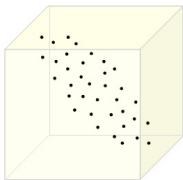


The major effect of a transformation matrix is determined by the eigenvector with the largest eigenvalue.

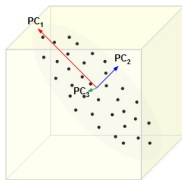
Eigenvectors as Principal Components



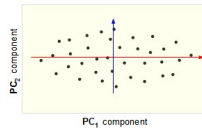
In *Principal Component Analysis (PCA)* or *Factor Analysis*, the Principal Components or Factors are the eigenvectors of a correlation or covariance matrix. The eigenvalues are the “factor loadings”.



a



b



c

Complex numbers



Complex numbers

$$ax^2 + bx + c = 0$$

Solve using the 'abc'-formula:

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-b \pm \sqrt{D}}{2a}$$

where $D = b^2 - 4ac$

What if $D < 0$?

The basic complex number

i is the basic complex number,
just as 1 is the basic real number.

It is defined as: $i^2 = -1$

which you can also write as: $i = \sqrt{-1}$

Working with complex numbers

$$\lambda^2 + 2\lambda + 2 = 0$$

$$\lambda_{1,2} = \frac{-2 \pm \sqrt{D}}{2} \quad \text{where} \quad D = 2^2 - 4 \times 1 \times 2 = 4 - 8 = -4$$

$$i^2 = -1 \Leftrightarrow i = \sqrt{-1}$$

$$\sqrt{D} = \sqrt{-4} = \sqrt{4}\sqrt{-1} = 2i$$

$$\lambda_{1,2} = \frac{-2 \pm 2i}{2} = -1 \pm i$$

$(-1 + i)$ and $(-1 - i)$ are called **complex conjugates**.

General complex numbers

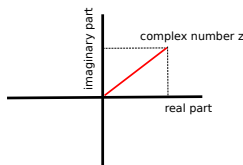
Complex number: $z = \alpha \pm i\beta$

Real part: $\text{Re}z = \alpha$

Imaginary part: $\text{Im}z = i\beta$

Can be displayed on the complex plane:

The complex plane



Looks similar to a 2D vector $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$.

Absolute value or modulus: $|z| = \sqrt{\alpha^2 + \beta^2}$

Calculus with complex numbers

Very similar to working with vectors:

Addition of two complex numbers:

$$z_1 = 3 + 10i, z_2 = -5 + 4i$$

$$z_1 + z_2 = (3 + 10i) + (-5 + 4i) = 3 - 5 + 10i + 4i = -2 + 14i$$

$$\begin{pmatrix} 3 \\ 10 \end{pmatrix} + \begin{pmatrix} -5 \\ 4 \end{pmatrix} = \begin{pmatrix} -2 \\ 14 \end{pmatrix}$$

Multiplication by real number:

$$z_1 = 3 + 10i$$

$$10 \times z_1 = 10(3 + 10i) = 30 + 100i$$

$$10 \begin{pmatrix} 3 \\ 10 \end{pmatrix} = \begin{pmatrix} 30 \\ 100 \end{pmatrix}$$

Calculus with complex numbers

Dissimilar to working with vectors:

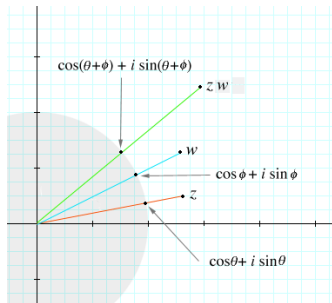
Multiplication of two complex numbers:

$$z_1 = 3 + 10i, z_2 = -5 + 4i$$

$$z_1 \times z_2$$

$$\begin{aligned} & (3 + 10i)(-5 + 4i) \\ & 3 \times (-5) + 3 \times 4i + 10i \times (-5) + 10i \times 4i \\ & -15 + 12i - 50i + 40i^2 \\ & -15 + 38i + 40i^2 \text{ and } i^2 = -1 \\ & -15 + 38i - 40 \\ & -55 - 38i \end{aligned}$$

Multiplication of two complex numbers



Calculus with complex numbers

Dissimilar to working with vectors:

Division of two complex numbers:

$$z_1 = 3 + 10i, z_2 = -5 + 4i$$

$\frac{z_1}{z_2} = \frac{z_1 \bar{z}_2}{z_2 \bar{z}_2}$, where $\bar{z}_2$ is the complex conjugate of z_2 .

$$\frac{(3 + 10i)(-5 - 4i)}{(-5 + 4i)(-5 - 4i)} = \frac{-15 - 40i - 50i - 40i^2}{25 + 20i - 20i + 16i^2}$$

$$\frac{-15 - 90i - 40i^2}{25 + 16i^2} \text{ and } i^2 = -1$$

$$\frac{-15 - 90i + 40}{25 - 16} = \frac{25 - 90i}{9} = \frac{25}{9} - 10i$$

Summary: Linear Algebra and Complex Numbers

- Working with scalars, vectors, matrices
- Matrix transformation: $A\vec{v} = \vec{w}$
- Determinant: $\det \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - cb$
- Eigenvectors $\mathbf{v}$ and eigenvalues λ : $A\mathbf{v} = \lambda\mathbf{v}$
- Finding eigenvalues of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$: $\det \begin{vmatrix} a - \lambda & b \\ c & d - \lambda \end{vmatrix} = 0$
- Finding eigenvectors:
 $\mathbf{v}_{1,2} = \begin{pmatrix} -b \\ a - \lambda_{1,2} \end{pmatrix}$ or $\mathbf{v}_{1,2} = \begin{pmatrix} -d + \lambda_{1,2} \\ c \end{pmatrix}$
- Complex numbers: $i^2 = -1$, $i = \sqrt{-1}$ and $z = \alpha \pm i\beta$